

Sales Prediction Management System

Defense and User Guide

Mapped to the supervisor software block diagram

How each block works in the running software

One sentence

SPMS is a web shop system that records sales, cleans the history, forecasts future demand with Linear Regression and LSTM, shows charts and reports, and recommends whether to reorder stock.

How the diagram is built

The supervisor diagram has one title box, Sales Prediction Management System, then six modules side by side. Each module has four smaller boxes under it. The left menu of the software uses the same six names.

Diagram module	Menu in the software	Who can open it
1. User Management	Dashboard, Manage Users, Logout	All users; Manage Users is admin only
2. Sales Data Management	Sales Entry, Sales History, Products, Customers	Staff, manager, admin
3. Data Processing and Cleaning	Clean and Prepare	Manager and admin
4. Sales Prediction and Forecasting	Predict and Evaluate	Manager and admin
5. Reports and Visualization	Charts and Reports	Manager and admin
6. Decision Support and Inventory Planning	Inventory Planning	Manager and admin

How to start the software

- Unzip or clone the project. Do not install PostgreSQL. The database is MySQL inside XAMPP.
- Once: double-click 1-SETUP.bat and wait until it says setup finished.
- Every time: double-click 2-START-SERVERS.bat, then open <http://localhost/spms/>
- If Forecasts says the engine is offline, wait about 10 seconds and refresh. Flask is still loading.
- Optional public link: double-click 3-START-NGROK.bat, open <http://127.0.0.1:4040>, copy the https address and add /spms/
- When finished: double-click STOP-SERVERS.bat.
- Password for every demo account is password123.

Role	Email	Job in the shop
Admin	admin@spms.local	Full control, users, and model retrain
Manager	manager@spms.local	Reports, forecasts, stock decisions
Staff	staff1@spms.local	Record sales and print receipts

Block 1 - User Management

This block controls who can enter the system and what they are allowed to touch. Passwords are stored as hashes, not plain text. After login, a session starts. Logout destroys that session.

Diagram box	What it means	Where in the software
Login / Authentication	Check email and password before opening the shop	Login page
Access Control	Staff, manager, and admin see different menus	Every page checks the role
Manage Users	Create accounts, change role, activate or deactivate	Manage Users (admin only)
Logout	End the session safely	Logout button at the top right

Defense line: staff can sell, managers can analyse, and only the admin can create logins or retrain the model.

Block 2 - Sales Data Management

This block is the daily shop work: record what was sold, keep product and customer records, and correct a sale if it was entered wrong.

Diagram box	What it means	Where in the software
Sales Data Entry	Record a sale: product, customer, quantity, date	Sales Entry
Product Data Management	Name, price, stock, reorder level, supplier	Products
Customer Data Management	Name, phone, email, address	Customers
Data Updates	Change a saved sale and adjust stock	Sales History, then Update

How a sale works

- The user picks a product. The screen shows the price and the stock.
- Quantity times unit price becomes the total.
- If quantity is more than stock, the sale is blocked. That stops overselling.
- If stock is enough, the sale is saved and stock goes down by that quantity.
- A receipt can be opened and printed.
- Update on Sales History puts the old quantity back into stock, then deducts the new quantity.

Suppliers and Promotions also live in this area. Promotions later become an external factor for forecasting.

Block 3 - Data Processing and Cleaning

Raw sales are messy for a model. Some days have many receipts. Some days have none. This block turns that mess into one clean number per day, plus extra clues. Open Clean and Prepare, pick a product, and choose how many days to look at.

Diagram box	What the software does	What you see
Data Validation	Drops rows with no date or quantity of zero	Invalid rows skipped count
Handle Missing Data	Days with no sale become quantity 0	Yellow rows marked Missing filled
Data Cleaning	Many sales on one day are added into one daily total	Clean qty column
Data Preparation	Adds weekday, weekend, month, and promo flag	Those columns on the prepared table

Why this matters: Linear Regression and LSTM both read this prepared series. If missing days were left as gaps, the timeline would be broken.

Block 4 - Sales Prediction and Forecasting

This block learns from the cleaned history and guesses future daily sales. Open Predict and Evaluate. Two models are used on purpose.

Model	Plain meaning	Role in the project
Linear Regression	A simple weighted formula from past clues to the next quantity	Baseline, easy to explain
LSTM	A neural network that reads sales as a sequence and keeps a memory of recent days	Main forecasting model

Diagram box	What it means	Where in the software
Model Training	Learn weights from past days and save the model	Admin button Update model, or python train.py
Generate Predictions	Ask for the next N days for one product	Run prediction
Forecast Results	Show the guessed quantities	Orange dashed line and the saved forecast table
Model Evaluation	Score how wrong each model was	MAE and RMSE table, Better MAE badge

How to read the chart

- Blue line = real past sales.
- Orange dashed line = the model's guess for future days. It is not money already earned.
- Horizon 7 means the next 7 days.
- Predicted quantity is always units of the selected product, for example tissue packs or cartons.

MAE and RMSE

- MAE is the average size of the mistake, in product units. Wrong by about 2 packs a day means MAE near 2.
- RMSE is similar, but a few very large mistakes pull it up more.
- Lower is better. They are not order quantities and they are not stock.
- Recent model training scores on the right lists the latest scores. The comparison table is for the product you selected.

Block 5 - Reports and Visualization

This block is for looking backward: what already sold, which products did well, and the shape of the trend. Open Charts and Reports. Pick a start date and an end date.

Diagram box	What you see	How to use it in defense
Charts and Graphs	Bar chart of daily revenue	A picture of sales over the chosen period
Trends Analysis	Daily revenue chart and sales by product	Which days and products are stronger
Performance Reports	Daily summary and monthly overview	Transactions, units, and revenue
Export Reports	Export CSV button	Open the file in Excel

The dashboard also shows today's revenue, this month, a 14-day chart, and low-stock alerts. That is a quick view. Charts and Reports is the fuller view.

Block 6 - Decision Support and Inventory Planning

Forecasts are only useful if a manager can act on them. Inventory Planning compares what is on the shelf with what the model thinks customers will buy.

Diagram box	What it means	Where in the software
Sales Insights	What sold recently, next to the forecast	Sold last 7 days column
Inventory Planning	Stock and reorder level versus predicted demand	Stock, reorder level, predicted demand
Business Decision Support	A plain sentence the manager can follow	Recommendation column
Recommendation Generation	Order a number of units, or do not order	Badge: Order X, Stock OK, or Need forecast

Example to memorise

Stock now is 20 units. Predicted demand for the next 7 days is 35 units. 35 is expected sales, not the stock you will have. 20 is not enough for 35, so the recommendation is to order about 15 units. If stock were 100 and demand were 35, the recommendation would be Stock OK.

Reorder level is a fixed warning the manager typed on the product, for example 10. It is separate from the forecast. A product can be above the reorder level and still need an order if the forecast is high.

How the blocks connect

- User Management lets the right person in.
- Sales Data Management stores products, customers, and each sale, and reduces stock.
- Data Processing and Cleaning builds a daily series and features.
- Sales Prediction and Forecasting trains Linear Regression and LSTM, then writes future quantities.
- Reports and Visualization shows what already happened.
- Decision Support compares the forecast with stock and says whether to order.

Browser to PHP pages to MySQL. The prediction pages call a Python Flask service on this same computer, port 5000. PHP never contains the neural network. That split matches a modular design: shop, database, and prediction engine.

Five-minute demo

- Login as staff1@spms.local. Open Sales Entry. Sell 1 unit. Open the receipt. Stock has gone down.
- Logout. Login as manager@spms.local. Open Dashboard. Show today's money and low stock.
- Open Clean and Prepare. Show a yellow missing day filled with 0, and the promo or weekend columns.
- Open Predict and Evaluate. Run prediction for 7 days. Blue is history. Orange is the guess. Show MAE.
- Open Charts and Reports. Show the trend chart and mention Export CSV.
- Open Inventory Planning. Read one recommendation out loud: Order X or Stock OK.
- Optional: login as admin and show Manage Users and the Update model button.

Questions a supervisor may ask

Q: What does the system do?

A: It records sales and forecasts future product demand so stock can be planned.

Q: Why are there three roles?

A: Staff sell. Managers analyse and plan stock. Admin controls users and retraining.

Q: Which diagram block is the forecast?

A: Block 4, Sales Prediction and Forecasting. Results are used again in Block 6.

Q: Why two models?

A: Linear Regression is the simple baseline. LSTM reads the sales timeline. We compare them with MAE and RMSE.

Q: What is predicted demand?

A: The sum of predicted units over the horizon, for example 7 days. It is expected sales, not future stock.

Q: What is a recommendation?

A: If predicted demand is greater than stock, order the difference. Otherwise stock is enough.

Q: Where is data cleaning?

A: Clean and Prepare. Missing days become 0. Same-day sales are added together. Features are added.

Q: Is the forecast guaranteed?

A: No. It supports a decision. A sudden event that is not in the history can still surprise the shop.

Q: What database is used?

A: MySQL through XAMPP. Not PostgreSQL.

Q: What if Forecasts says the engine is offline?

A: Flask is not ready. Wait and refresh, or run 2-START-SERVERS.bat again.

Limits to say honestly

- The model needs enough past sales. A brand-new product with no history cannot be forecast until it is trained.
- Promo helps only if the promotion was entered in the Promotions page.
- A public ngrok link works only while that computer stays on.
- MAE and RMSE describe past test error. They do not promise next week will match exactly.

Read Blocks 2, 4, and 6 again before the defense, then practise the five-minute demo once.